The Maximal Inequality

Wednesday, November 12, 2025 6:07 PM

Definition: $\{X_n\}$ is said to be a supermartingale w.r.t. $\{Y_n\}$ if $E(|X_k|) < \infty \ \forall k$ and

$$E(X_{k+1} \mid Y_k, \ldots, Y_0) \leq X_k \qquad \text{a}$$

Definition: A r.v. T is said to be a stopping time w.r.t. a random process $X_0, X_1, X_2, \ldots$ if the event $\{T = k\}$ is determined by $X_0, X_1, X_2, \ldots, X_k$. In other words,

$$I_{\{T=k\}} = f(X_0, \ldots, X_k) \quad \text{w.p.1}$$

(f: some deterministic function)

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Example: Let $S_n = X_1 + \cdots + X_n$, where $X_i = \text{Rad}(\frac{1}{2})$ indep. variables, i.e.,

$$X_i = \begin{cases} +1 & \text{w.p. } \frac{1}{2} \\ -1 & \text{w.p. } \frac{1}{2} \end{cases}$$

$$T = \min\{k \geq 1 : S_k = 10\} \quad \left(\begin{array}{l}\text{first time to} \\ \text{hit } 10\end{array}\right)$$

Clearly,

$$\{T=k\} = \{X_1+\cdots+X_i<10 \ \forall i,\ X_1+\cdots+X_k=10\}$$

Thus, T is a stopping time.

Example:

$$T = \max\{k\le 100 : S_k=10\}$$

(last time to hit 10 before $k=100$)

Clearly $\{T=50\}$ depends on $X_{51}, X_{52}, \ldots, X_{100}$

so T is not a stopping time. □

Optional stopping theorem: Let X be a supermartingale w.r.t Y and S, T be stopping times w.r.t. Y s.t.

$$S \le T \le N \quad \text{w.p.1 for some finite } N.$$

Then, $$E(X_T \mid X_S) \le X_S$$

Proof: Let

$$Z_k = X_{k\wedge T} = \begin{cases} X_k & \text{if } k\le T \\ X_T & \text{if } k>T \end{cases}$$

We now claim that Y_k is also a supermartingale.

supermartingale.

$$E(Z_k | Y_0, \ldots, Y_{k-1}) = E(X_{k \wedge T} | Y_0, \ldots, Y_{k-1})$$

$$= E(X_T I_{k \le T} + X_k I_{k > T} | Y_0, \ldots, Y_{k-1})$$

$$\le X_T I_{k \le T} + X_{k-1}$$

Since $T \le N$,

$$Z_N = X_T \quad \text{and} \quad Z_S = X_S$$

$$\Rightarrow E(Z_N | Y_0, \ldots, Y_S) \le Z_S$$

$$\Rightarrow E(X_T | Y_0, \ldots, Y_S) \le X_S$$

□

The Maximal Inequality (Doob)

Let X be a non-negative supermartingale w.r.t. Y. Then, for any $\lambda > 0$

$$P\left(\max_{0 \le k \le n} X_k \ge \lambda \right) \le \frac{E(X_0)}{\lambda}$$

Proof: Let $\tau = \min \{ k \in \{0, 1, \ldots, n\} : X_k \ge \lambda \}$ with the convention $\inf \phi = \infty$. So $\tau \wedge n$ is a bounded stopping time.

$\Rightarrow$ by the optional stopping theorem,

$$E(X_{\tau\wedge n}) \leq E(X_0)$$

On the event $\{\tau \leq n\}$, $X_{\tau\wedge n} = X_\tau \geq \lambda$ by the definition of τ.

On the event $\{\tau > n\}$, $X_{\tau\wedge n} = X_n \geq 0$ by assumption.

$$\Rightarrow \quad X_{\tau\wedge n} \geq \lambda \, I_{\tau \leq n}$$

$$\Rightarrow \quad E(X_{\tau\wedge n}) \geq \lambda \, P(\tau \leq n)$$

$$= \lambda \, P\left(\max_{0 \leq k \leq n} X_k \geq \lambda \right)$$

$$\Rightarrow P\left(\max_{0 \leq k \leq n} X_k \geq \lambda \right) \leq \frac{E(X_{\tau\wedge n})}{\lambda}$$

$$\leq \frac{E(X_0)}{\lambda}$$

□

Application to concentration

Let $X_1, X_2, \ldots$ be sub-Gaussian i.i.d. r.v.'s, with

$$\text{ex: } 1 \qquad \theta^2 \sigma^2$$

with

$$E\left(e^{\theta X_i}\right) \le e^{\frac{\theta^2\sigma^2}{2}} \quad \forall \theta \in \mathbb{R}$$

Also, assume $E(X_i) = 0$. Let

$$S_n = X_1 + \cdots + X_n .$$

We are interested in upper bounding

$$P\left(\max_{1 \le k \le n} S_k \ge t \right) \le ?$$

Let $M_k = \exp\left(\theta S_k - \frac{k\theta^2\sigma^2}{2} \right)$, $k = 0, 1, \ldots, n$

$$E\left(M_k \mid X_1, \ldots, X_{k-1} \right) = \exp\left(\theta S_{k-1} - \frac{k\theta^2\sigma^2}{2} \right) E\left(e^{\theta X_k}\right)$$

$$\le e^{\theta S_{k-1} - \frac{k\theta^2\sigma^2}{2}} \; e^{\frac{\theta^2\sigma^2}{2}}$$

$$= \exp\left(\theta S_{k-1} - (k-1)\frac{\theta^2\sigma^2}{2} \right) = M_{k-1}$$

and $E(M_1) = \exp\left(\theta S_1 - \frac{\theta^2\sigma^2}{2} \right) \le 1 = E(M_0)$

$\Rightarrow$ $\{M_k\}_{k=0}^n$ is a supermartingale.

Next, note that

$\max_{1 \le k \le n} S_n \ge t$ means $\exists$ some $k \in \{1, \ldots, n\}$,

where $S_k \geq t$

$$\Rightarrow \quad \exists\, k \in \{1, \ldots, n\} \text{ s.t. } \quad M_k \geq \exp\left(\theta t - \frac{k\theta^2\sigma^2}{2}\right)$$

$$\geq \exp\left(\theta t - \frac{n\theta^2\sigma^2}{2}\right)$$

$$\Rightarrow \left\{ \max_{1 \leq k \leq n} S_n \geq t \right\} \subseteq \left\{ \max_{1 \leq k \leq n} M_k \geq \exp\left(\theta t - \frac{n\theta^2\sigma^2}{2}\right) \right\}$$

Apply the maximal inequality:

$$P\left(\max_{1 \leq k \leq n} S_n \geq t \right) \leq P\left(\max_{1 \leq k \leq n} M_k \geq \exp\left(\theta t - \frac{n\theta^2\sigma^2}{2}\right) \right)$$

$$\leq P\left(\max_{0 \leq k \leq n} M_k \geq \exp\left(\theta t - \frac{n\theta^2\sigma^2}{2}\right) \right)$$

$$\leq \frac{E(M_0)}{\exp\left(\theta t - \frac{n\theta^2\sigma^2}{2}\right)}$$

$$= \frac{1}{\exp\left(\theta t - \frac{n\theta^2\sigma^2}{2}\right)}$$

Optimizing over θ:

$$t - \frac{n\theta\sigma^2}{2} \Rightarrow \quad \theta = \frac{t}{n\sigma^2}$$

$t - \frac{n\theta\sigma^2}{2} \Rightarrow \theta = \frac{t}{n\sigma^2}$

$$\Rightarrow \boxed{P\left(\max_{1\le k\le n} S_k \ge t\right) \le \exp\left(\frac{-t^2}{2n\sigma^2}\right)}$$